

SET 9 – ONLINE LEARNING ACTIVITY ANALYSIS

DSA0601 – DATA HANDLING AND VISUALIZATION

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Course	Data Handling and Visualization
Set	9

Dataset: Online_Learning_Activity.csv

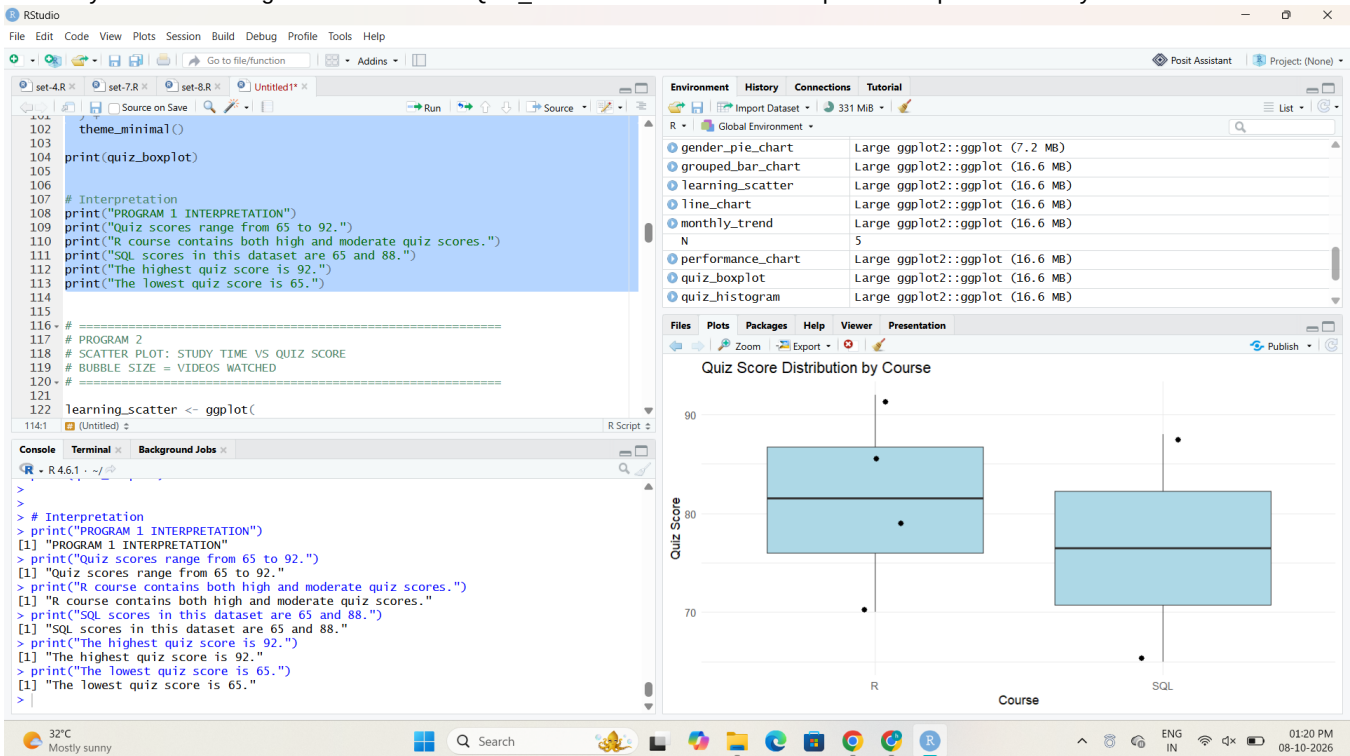
Student ID	Gender	Age	Course	Study Time	Videos	Quiz Score	Login Date
L01	Male	20	R	3.5	12	78	2025-01-05
L02	Female	22	R	4.2	15	85	2025-01-05
L03	Male	19	SQL	2.0	8	65	2025-02-08
L04	Female	21	R	5.0	18	92	2025-02-08
L05	Male	23	R	2.5	9	70	2025-03-12
L06	Female	20	SQL	4.0	14	88	2025-03-12

Objectives

- Plot a histogram and boxplot to study Quiz_Score distribution and student performance.
- Create a bubble scatter plot between Study_Time and Quiz_Score using Videos_Watched as bubble size.
- Convert Login_Date to Date format, calculate monthly average quiz score and apply moving-average smoothing.
- Create a Tableau mini dashboard with course averages, scatter plot and Gender/Course filters.

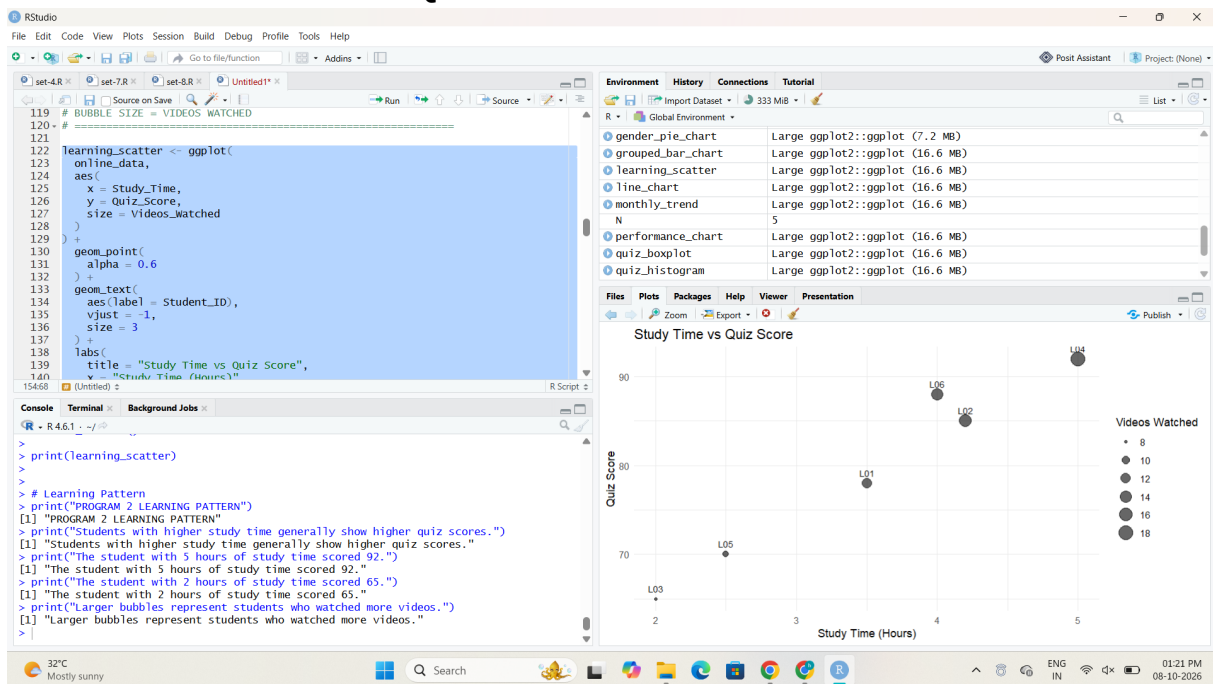
PROGRAM 1 – QUIZ SCORE DISTRIBUTION

The R analysis uses a histogram for the overall Quiz_Score distribution and a boxplot to compare scores by Course.



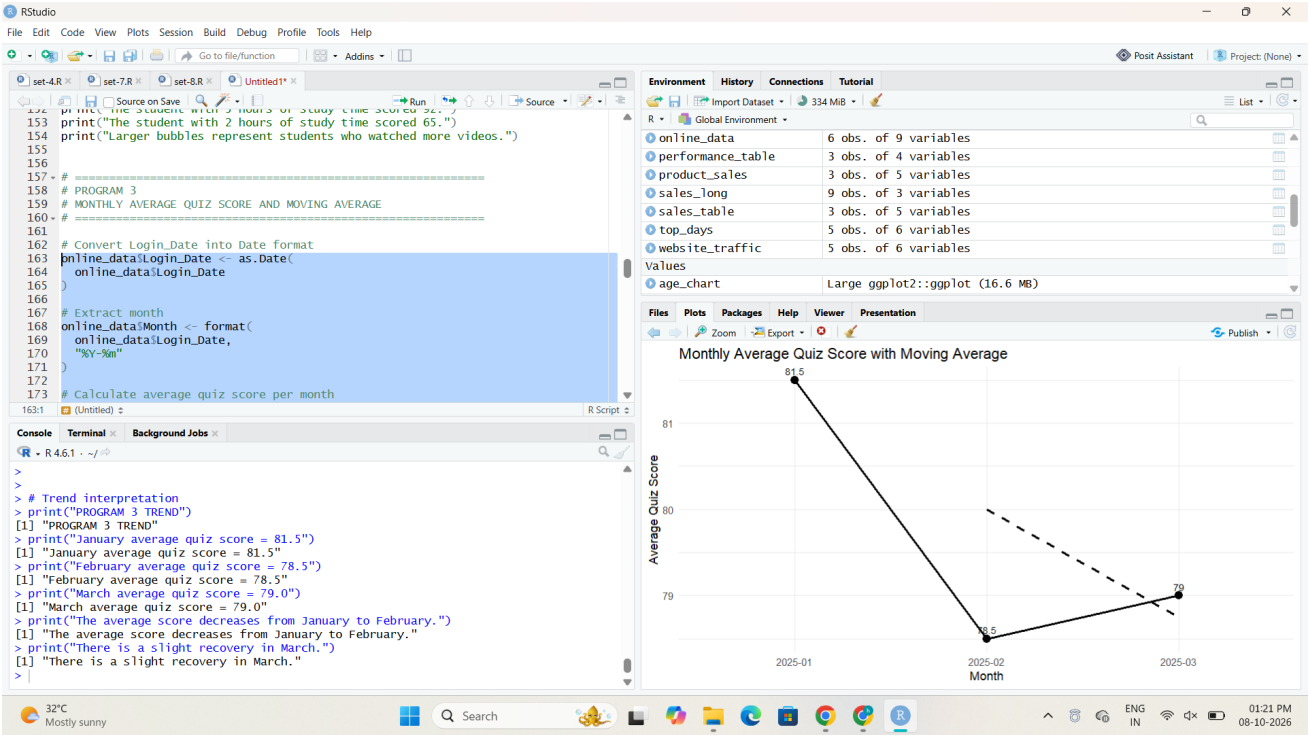
Interpretation: Scores range from 65 to 92. The R course has scores from moderate to high levels, while SQL has scores of 65 and 88. The highest score is 92 and the lowest is 65.

PROGRAM 2 – STUDY TIME VS QUIZ SCORE



Learning pattern: the supplied observations generally show higher quiz scores with greater study time. Bubble size represents Videos_Watched, and transparency makes overlapping observations easier to see.

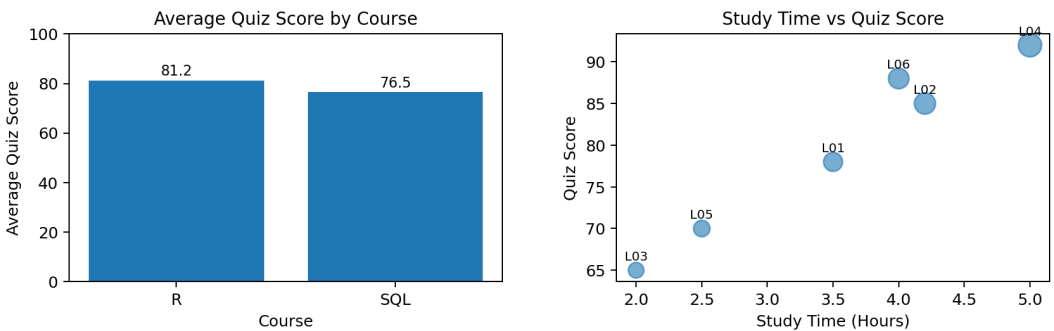
PROGRAM 3 – MONTHLY AVERAGE QUIZ SCORE WITH MOVING AVERAGE



Monthly averages: January = 81.5, February = 78.5, March = 79.0. The trend decreases from January to February and shows a slight recovery in March. The dashed line represents the moving-average smoothing.

PROGRAM 4 – TABLEAU MINI DASHBOARD

Online Learning Activity Dashboard

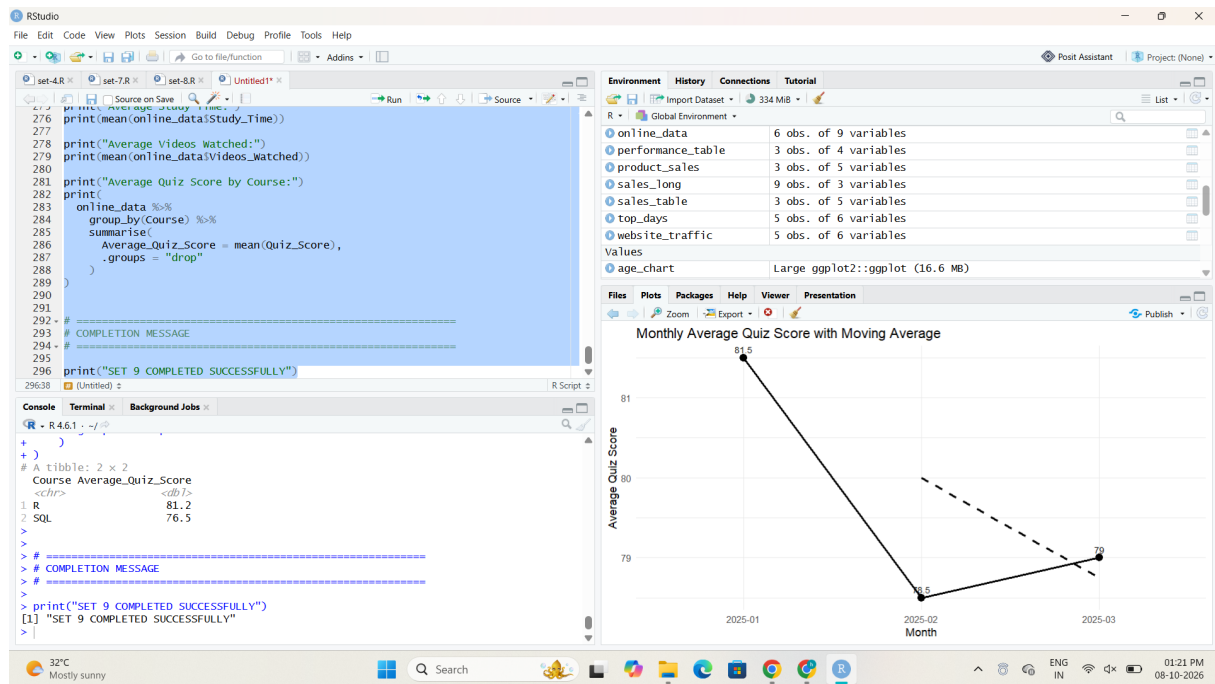


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Illustrative Tableau-style dashboard based on the supplied dataset

The dashboard combines average quiz score by Course, Study Time vs Quiz Score, and the complete activity table. Gender and Course can be added as interactive filters in Tableau.

RESULT AND ANALYSIS



Key Results

- Overall average Quiz Score = 79.83.
- Average Study Time = 3.53 hours.
- Average Videos Watched = 12.67.
- Average Quiz Score by Course: R = 81.25; SQL = 76.50.
- Monthly average scores are 81.5, 78.5 and 79.0 from January to March.

Conclusion

The Online Learning Activity dataset was successfully analyzed using histogram, boxplot, bubble scatter plot and monthly trend visualization. The results provide a clear view of quiz performance, study time, video engagement and monthly changes, while the Tableau mini dashboard supports interactive exploration.